

Amir Mehdi Parviz

AI Engineer | LLM Systems, Document AI & AI Backend

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Professional Summary

AI Engineer with 3+ years of hands-on experience building production-oriented systems across LLM applications, Persian OCR and document intelligence, computer vision, model fine-tuning/evaluation, and AI backend services. Strong in Python, PyTorch, FastAPI, Docker/Linux, GPU inference, REST APIs, and PostgreSQL. Experienced in taking models from data preparation and experimentation through validation, API integration, self-hosted serving, and operational workflows.

Core Skills

Languages & Backend: Python, FastAPI, REST APIs, SQL, PostgreSQL, SQLAlchemy, Alembic, Redis, Celery, JWT authentication, Pytest

AI & ML: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, PEFT/LoRA, LLMs, VLMs, LangChain, MCP (Model Context Protocol), NLP, time-series analysis

Document AI & Computer Vision: Persian OCR, structured extraction, document understanding, OpenCV, Tesseract/PyTesseract, YOLOv8, object detection, image preprocessing

Model Serving & Infrastructure: vLLM, OpenAI-compatible APIs, Docker, Docker Compose, Linux, systemd, tmux, CUDA/GPU inference, quantization, model optimization

Data & Evaluation: Dataset preparation, synthetic data generation, feature engineering, model evaluation, validation design, human-review workflows, structured JSON outputs

Engineering: Git/GitHub, modular service design, API integration, deployment workflows, testing, technical documentation

Professional Experience

Vira RNDlab | Iran | AI Engineer (Part-time / Project-based) | 2026 - Present

- Contributed to project-based AI systems spanning document AI/OCR, LLM/VLM applications, model serving, evaluation, and AI backend integration.
- Built and maintained self-hosted model-serving workflows using vLLM, Linux, CUDA/GPU inference, and OpenAI-compatible APIs.
- Implemented model evaluation, fine-tuning, validation, and human-review workflows for production-oriented AI projects.
- Developed Python/FastAPI service layers and integrated AI models with application and product workflows.
- Worked on model selection, deployment, performance optimization, and reliable AI system integration across project engagements.

Caitech | Iran | AI Developer & Data Analyst | Jun 2020 - 2025

- Developed AI-powered applications across LLM/NLP, document processing, computer vision, data analysis, API integration, and product delivery.
- Built chatbot and support workflows combining web search, document context, and OpenAI-compatible LLM APIs.
- Developed computer-vision pipelines for object detection, lane detection, and pedestrian tracking in autonomous-vehicle scenarios.
- Performed EEG/time-series research and analysis and collaborated with software/ML teams to integrate models into usable services and validate behavior with real data.

Selected Projects

Persian OCR & Document Intelligence Platform

- Built a Persian document pipeline with image preprocessing, full-page tiling, OCR/VLM extraction, structured JSON output, field validation, and review workflows. Tech: Python, OpenCV, Tesseract, FastAPI, Qwen/Qwen-VL, vLLM, CUDA, Linux.

Persian V-OCR Model Training & Evaluation

- Prepared synthetic and real datasets, implemented LoRA fine-tuning, preserved base/candidate adapters, and designed frozen validation and human-review checks. Tech: PyTorch, Transformers, Qwen3-VL, PEFT/LoRA, BF16, CUDA.

OPRS / Smart Survey System

- Developed a smart survey system for structured questionnaire workflows, response collection, and AI-assisted processing within an application-oriented architecture.

Speech-enabled AI Response System (STT + LLM + TTS)

- Built a voice-response pipeline that converts user speech to text, processes the request with an LLM, and synthesizes the generated response back to speech for interactive AI assistance.

AI Document Workflow / DABIRKHANEH

- Developed a Persian RTL AI workflow for letter classification, drafting, validation, and workflow status visibility using multiple AI roles, FastAPI, PostgreSQL, REST APIs, and OpenAI-compatible model services.

ECG P-Wave and AF Classification Pipeline

- Implemented a two-stage detection/classification workflow with annotated images, CSV/JSON outputs, HTML review pages, and RTL reporting. Tech: YOLOv8, OpenCV, scikit-learn, HistGradientBoostingClassifier.

Additional Engineering Projects

- Productized a club-management SaaS/desktop application and built containerized delivery workflows using Docker Compose; also developed a controlled AI intent-execution service for validated application operations.

Education

Kermanshah University of Technology | Iran | Bachelor of Chemical Engineering | 2022 - 2026